How reliable are IMOS Data?

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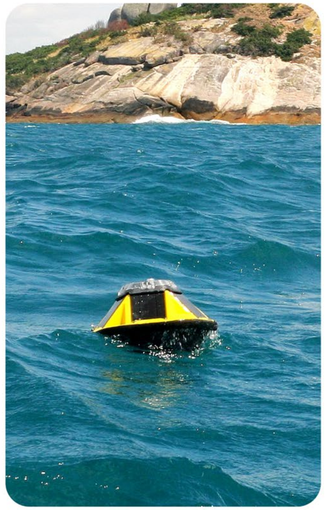
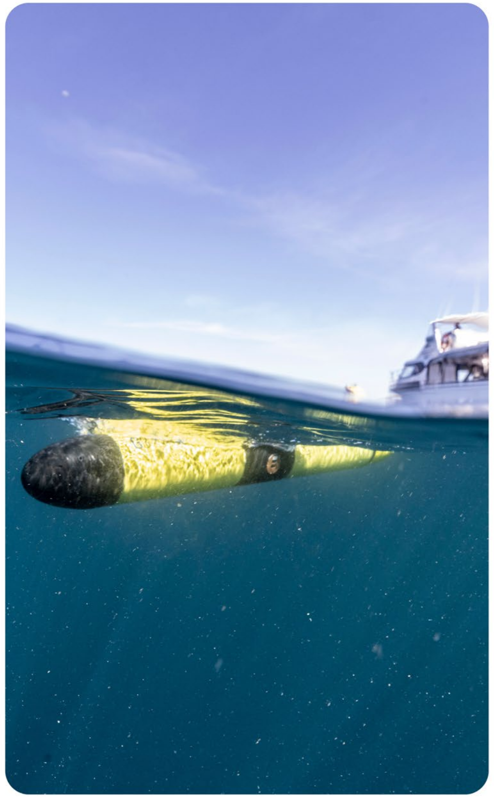
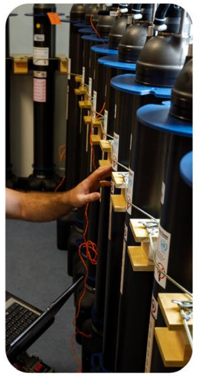
Chair, Ocean Best Practice System (OBPS)







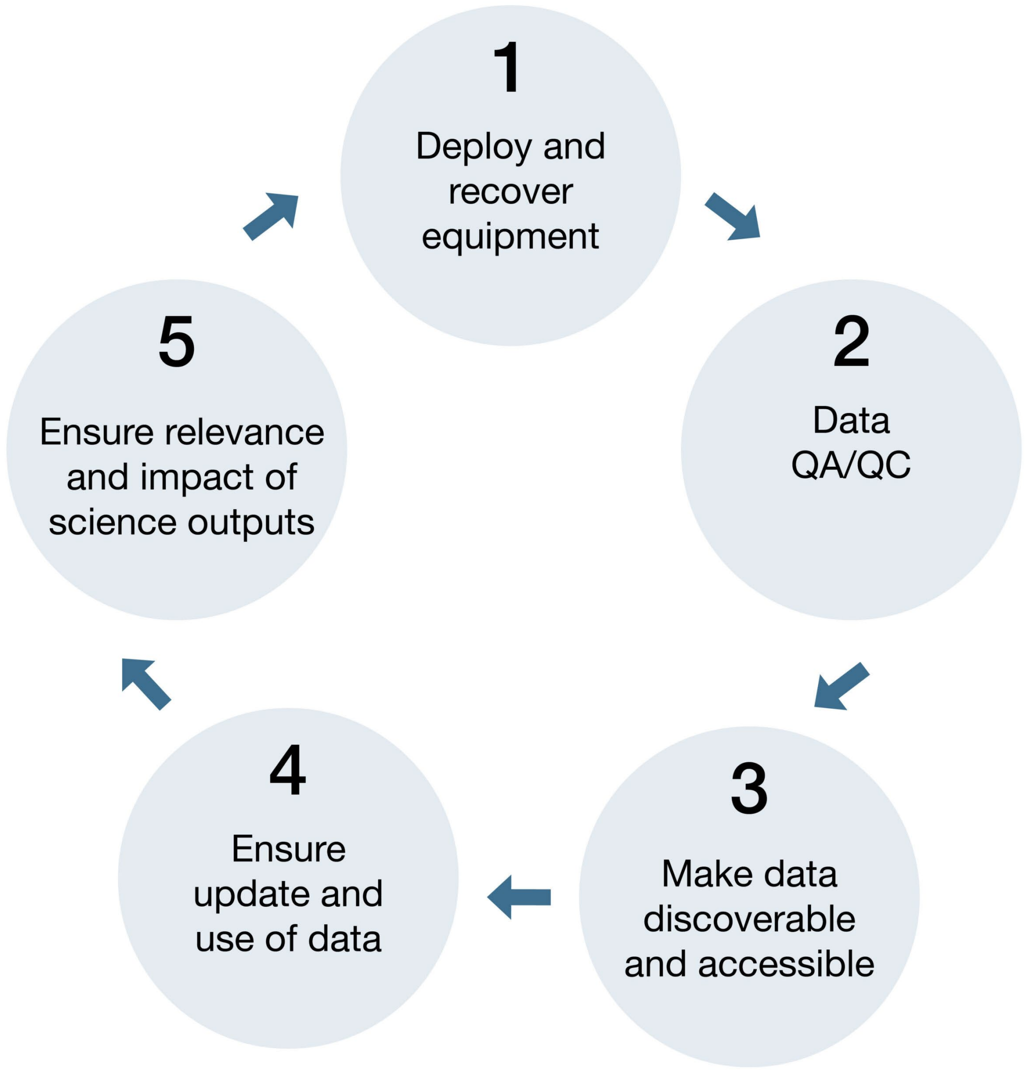
IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

Standardising Data Practices

Standardised SOPs, and QA/QC processes for the **collection, treatment, management, and delivery of IMOS data streams** by IMOS Facilities/sub-Facilities

Why this matters:

- Ensure the highest possible data quality and interoperability
- Improve transparency, traceability, and reproducibility
- Build user confidence in the quality and reliability of IMOS data
- Greater trust in data → Greater uptake → Greater impact
- Reassure funders that investments are generating meaningful impact



1
Deploy and recover equipment

2
Data QA/QC

Standardising Data Practices

All IMOS Facilities/sub-Facilities provide to the IMOS Office with **written documentation** covering:

- Calibration procedures
- Sample collection methods
- Storage protocols
- Sample processing and analysis workflows
- Data processing, management, and QA/QC procedures

This **applies to all variables collected and derived** by the Facilities/sub-Facilities.

All IMOS procedures and protocols are registered via:

- AODN Metadata Record
- Ocean Best Practices System (OBPS) Repository
- IMOS webpage

Maintain up-to-date documentation to build trust and ensure ongoing relevance and accuracy.

Best Practices in Ocean Observing

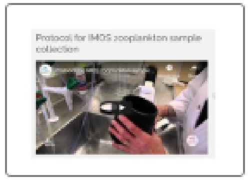
- **IMOS currently chairs the OBPS** and is a **recognized endorsing organisations for ocean best practices worldwide**
- **OBPS – IMOS Community Repository:** 63 Submissions including 56 reports and manuals, 7 videos

Videos (2016-2018)

Protocol for TSS Blank Collection. [Training video]

Integrated Marine Observing System (IMOS) (CSIRO/Integrated Marine Observing System (IMOS), Canberra, Australia, 2018)

Instructional video for Integrated Marine Observing System (IMOS) TSS Blank Collection – Protocol for IMOS TSS blank collection (2.08 mins) ....



Protocol for IMOS zooplankton sample collection. [Training video]

Integrated Marine Observing System (IMOS) (CSIRO/Integrated Marine Observing System (IMOS), Canberra, Australia, 2016)

Instructional video for Integrated Marine Observing System (IMOS) biogeochemical water sampling procedures – Protocol for IMOS zooplankton sample collection

Protocol for IMOS TSS sample collection. [Training video]

Integrated Marine Observing System (IMOS) (CSIRO/Integrated Marine Observing System (IMOS), Canberra, Australia, 2018)

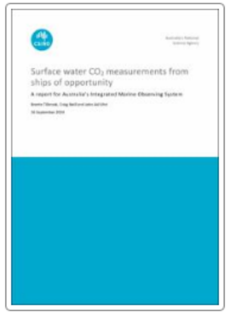
Instructional video for Integrated Marine Observing System (IMOS) biogeochemical water sampling procedures – Protocol for IMOS TSS sample collection (4.29 mins) ....

Documents (2002-2024)

Towards a best practice in acoustic telemetry mooring design.

Hartery, Cassandra; Bowen, Emma; Bate, Caitlin; McDowall, Phil; van den Broek, James; Whoriskey, Frederick; Jaine, Fabrice (Ocean Tracking Network (OTN) and Integrated Marine Observing System (IMOS), Dalhouse, Canada, 2025)

This document presents trialled and tested, best practice methods for the subsea deployment of acoustic telemetry equipment developed by two of the world's longest operating animal telemetry networks, the Ocean Tracking ...



Surface water CO2 measurements from ships of opportunity, A report for Australia's Integrated Marine Observing System.

Tilbrook, Bronte; Neill, Craig; Akl Uhri, John (CSIRO, Hobart, Australia, 2024)

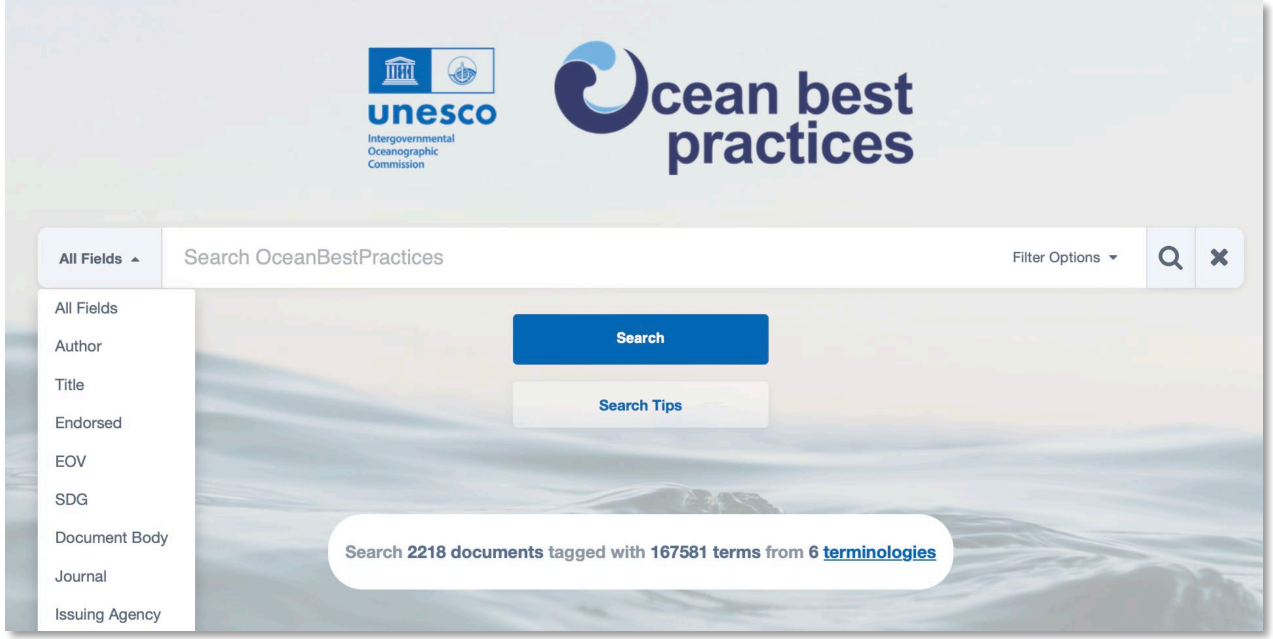
Underway measurements of the fugacity of carbon dioxide (fCO2) in surface waters and the atmosphere are made using General Oceanics Incorporated Model 8050 or 8060 systems (GO) equipped with LI-COR gas analysers.

IMOS Integrated Marine Observing System

Inclusion of a record in the repository does not indicate that it is a best practice.

The OBPS-IMOS community repository - Benefits

- **DOI and Metadata Generation**: Ensures citability and discoverability.
- **Wide Reach**: Broad audience with usage insights
- **Structured Categorization**: By maturity level, EOVs, endorsement etc. helping users find/apply the most relevant and well-developed practices.
- **Collaboration & Improvement**: Facilitates sharing, feedback, and continuous improvement/refinement - progressing from good-better-best.



Best Practice Manual for SMRU CTD Satellite Relay Data Loggers: Instrument Calibration, Near Real-Time and Delayed Mode Data QA/QC. Version 1.0.

View/Open
PDF (2.033Mb)

Date
2024

Author
Jonsen, Ian
McMahon, Clive
Harcourt, Rob

Status
Published

Pages
24pp.

Metadata
Show full item record

This document is the IMOS Animal Tracking Facility's Best Practice manual for near real-time and delayed-mode processing of physical and behavioural observations collected using Sea Mammal Research Unit CTD Satellite Relay Data loggers (SMRU CTD-SRDL). The Animal Tracking Facility deploys SMRU CTD-SRDL's on southern elephant seals (Mirounga leonina) and Weddell seals (Leptonychotes weddellii) in the Southern Ocean and on olive ridley sea turtles (Lepidochelys olivacea) and flatback sea turtles (Natator depressus) in the Timor and Arafura Seas. The data transmitted by these instrumented animals contributes to the study of ocean structure and dynamics by supplying temperature and salinity observations within the upper ocean in high latitude, shallow coastal and tropical regions that are historically under-sampled by traditional observing platforms. This document describes the calibration methods used by the Animal Tracking Facility prior to instrument deployment and the quality analys.....

Resource URL
https://catalogue-imos.aodn.org.au/geonetwork/srv/eng/catalog.search#/metadata/3e0148c8-295d-4804-94c3-4a80314ad746

Publisher
Integrated Marine Observing System (IMOS)
Hobart, Australia

Document Language
en

Sustainable Development Goals (SDG)
14.a

Essential Ocean Variables (EOV)
Subsurface temperature
Sea surface temperature
Sea surface salinity
Subsurface salinity
Ocean bottom pressure
Marine turtles, birds, mammals abundance and distribution

Maturity Level
Mature

Spatial Coverage
Southern Ocean
Timor Sea
Arafura Sea

DOI Original
https:doi.org/10.26198/ev75-0j83

Citation
Jonsen, I., McMahon, C. and Harcourt, R. (2024) Best Practice Manual for SMRU CTD Satellite Relay Data Loggers: Instrument Calibration, Near Real-Time and Delayed Mode Data QA/QC. Version 1.0. Hobart, Australia, Integrated Marine Observing System, 24pp. DOI: https:doi.org/10.26198/ev75-0j83).

URI
https://repository.oceanbestpractices.org/handle/11329/2571

Collections
IMOS Community Practices [55]

Data management in IMOS-AODN

1
Deploy and recover equipment

2
Data QA/QC

Standardised data formats:

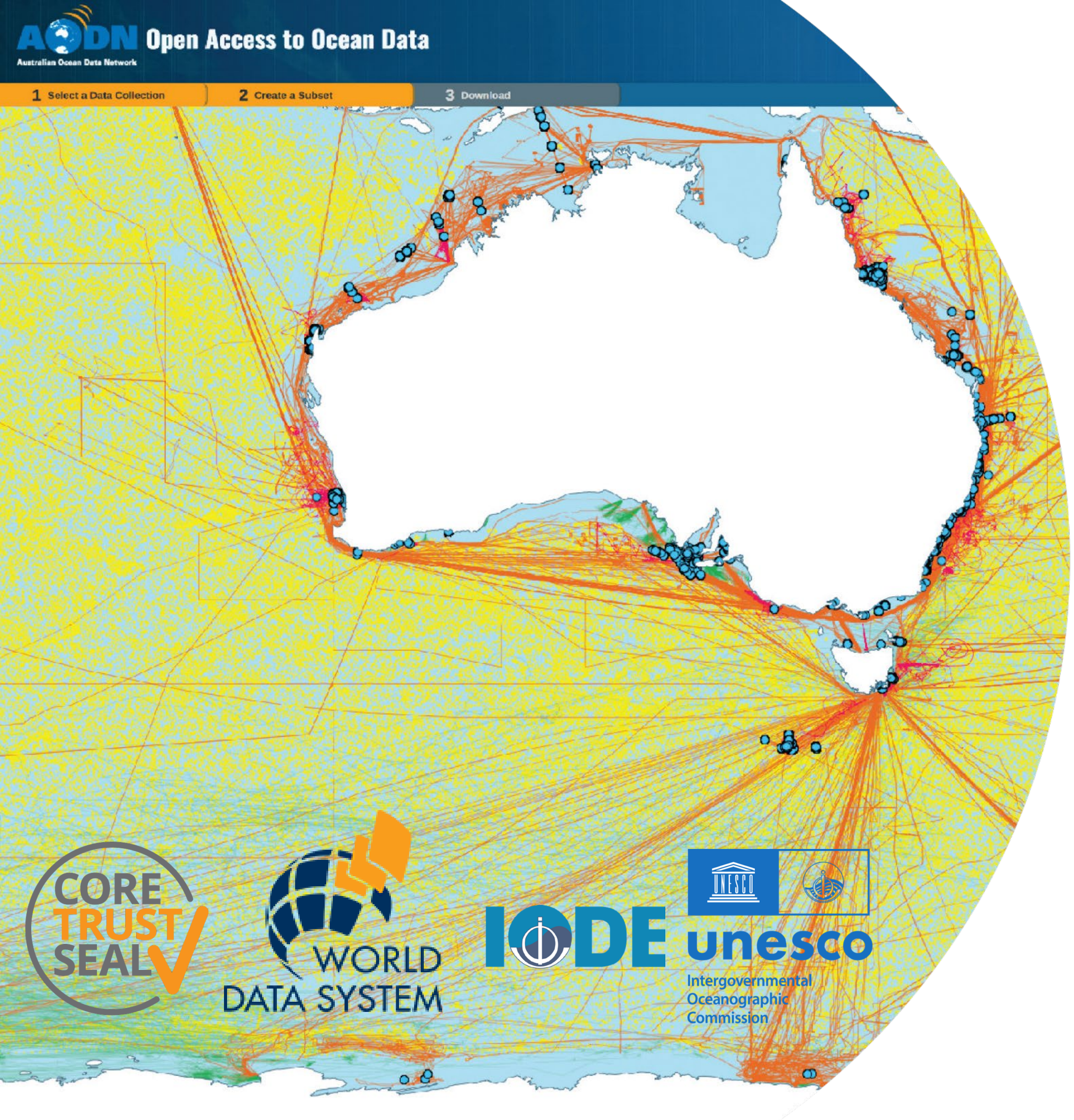
- IMOS-AODN developed a MATLAB toolbox for data processing
    - Process and export data from moorings and CTD profiles in the required IMOS format, ready to be ingested by AODN pipeline infrastructure.
- Data ingestion pipelines include a compliance checker (based on the US IOOS version) to ensure data adhere to Climate and Forecast (CF) and IMOS conventions.

Vocabularies:

- IMOS-AODN is an important partner in developing data vocabularies in Research Vocabularies Australia, in collaboration with the Australian Research Data Council (ARDC).
- IMOS both submits and re-uses content from the NERC (National Environmental Research Council) Vocabulary Server (NVS) which is operated by the British Oceanographic Data Centre.

Metadata

- IMOS-AODN has developed standardised descriptions of datasets (ISO 19115 compliant metadata)



3

Make data discoverable and accessible

Data Delivery

All data collected is made **publicly available through the Australian Ocean Data Network (AODN).**

- World Data System member
- Hold a Core Trust Seal certification
- Accredited International Oceanographic Data and Information Exchange (IODE) National Oceanographic Data Centre.
- IMOS data adhere to FAIR principles:
    - **F**indable
    - **A**ccessible (Open-Access)
    - **I**nteroperable
    - **R**eusable

4
Ensure update and use of data

5
Ensure relevance and impact of science outputs

Ensuring Data Relevance, Use and Impact

IMOS ensures that data collection **remains relevant, timely, and impactful** by aligning its observations and activities with:

National priorities, including:

- NMSC Decadal Strategy
- National Science Priorities
- Sustainable Ocean Plan
- National environmental reporting requirements
- Stakeholder, and rights holder need

International prioirties, such as:

- UN Ocean Decade Indicators
- Sustainable Development Goals (SDGs)
- Global Ocean Observing System (GOOS)
- International Quality-Controlled Ocean Database (IQuOD)
- US-IOOS Quality Assurance for Real-Time Oceanographic Data(QARTOD) project
- Etc.

4

Ensure update and use of data

5

Ensure relevance and impact of science outputs

Acknowledging IMOS

Users of IMOS data are required to clearly acknowledge the source material by including the following statements:

- **Long version:** *Data were sourced from Australia's Integrated Marine Observing System (IMOS) – IMOS is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent.*
- **Short version:** *Data were sourced from Australia's Integrated Marine Observing System (IMOS) – IMOS is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS).*

Requirements

- The statement above is required over and above the use of **in-text citations** (i.e., it **must be included in acknowledgements text**).
- Correct acknowledgement and citation **allow the IMOS Office to demonstrate impact, find uses, and increase recognition** of the vital place the observing system has in marine and climate science.

NCRIS
National Research Infrastructure for Australia
An Australian Government Initiative

Australia's Integrated Marine Observing System is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent.

PRINCIPAL PARTICIPANTS

- UNIVERSITY of TASMANIA (Lead Agent)
- Australian Government
- Australian Institute of Marine Science
- Australian Government Bureau of Meteorology
- CSIRO
- Government of South Australia
- SARDI South Australian Research and Development Institute
- The University of Western Australia
- sims sydney institute of marine science
- UTS
- The University of Sydney
- Macquarie University
- UNSW Sydney

SIMS is a partnership invoving four universities

ASSOCIATE PARTICIPANTS

- Curtin University
- Australian Government Department of Climate Change, Energy, the Environment and Water Australian Antarctic Division
- Australian Antarctic Program
- Deakin University
- The University of Melbourne

IMOS thanks the many other organisations who partner with IMOS, providing co-investment, funding and operational support, including investment from the Tasmanian and Western Australian Governments.